ClinicaAI — Methodology Report

ClinicaAI | October 2026

1. Educational need

ClinicaAI addresses a practical challenge in medical education: applying theoretical knowledge to clinical cases. An initial descriptive observation of 30 students found different proportions scoring above 10/20 in theory and clinical-case assessments (83% and 32%). This observation motivates further work but does not establish product efficacy.

2. Why generative AI?

Generative AI was selected to support flexible, natural and context-aware virtual-patient dialogue. The choice follows an initial exploratory comparison and remains open to structured evaluation.

3. Comparative approach

The AI Comparison Lab examines rule-based, semantic-matching and generative answers using a shared clinical dossier. The generative method uses Google Gemini. This makes it possible to study relevance, consistency and conversational quality across approaches.

4. Educational benchmark

The framework contains 100 questions, divided into Q01–Q50 for development and Q51–Q100 for evaluation. Relevant dimensions include question understanding, fidelity to the scenario, conversational naturalness and clinical coherence.

5. Medical review and validation approach

ClinicaAI includes a medical review portal where physicians can examine virtual-patient answers, assess their clinical relevance, and provide structured observations. Two physicians contributed feedback during the early evaluation stages.

The system also offers educational feedback on students' clinical reasoning, including an indicative score and personalized recommendations. This approach combines generative AI and medical expertise to progressively strengthen simulation quality and relevance.

6. Continuous improvement and future development

ClinicaAI follows an iterative development process focused on clinical quality, answer reliability and user experience.

Next steps include expanding medical scenarios, refining clinical-reasoning assessment, improving the consistency of generated answers and broadening evaluations by healthcare professionals. The goal is an interactive educational platform adaptable to multiple medical specialties with medical oversight and rigorous evaluation.

7. Development plan

Near-term priorities are to expand and refine cases and feedback, gather documented student usability feedback, and progressively broaden structured medical review.

8. Conclusion

ClinicaAI supports active learning through fictional clinical scenarios. It is an educational prototype: AI-generated answers and scores are indicative, not a substitute for clinical judgment or real-world care.